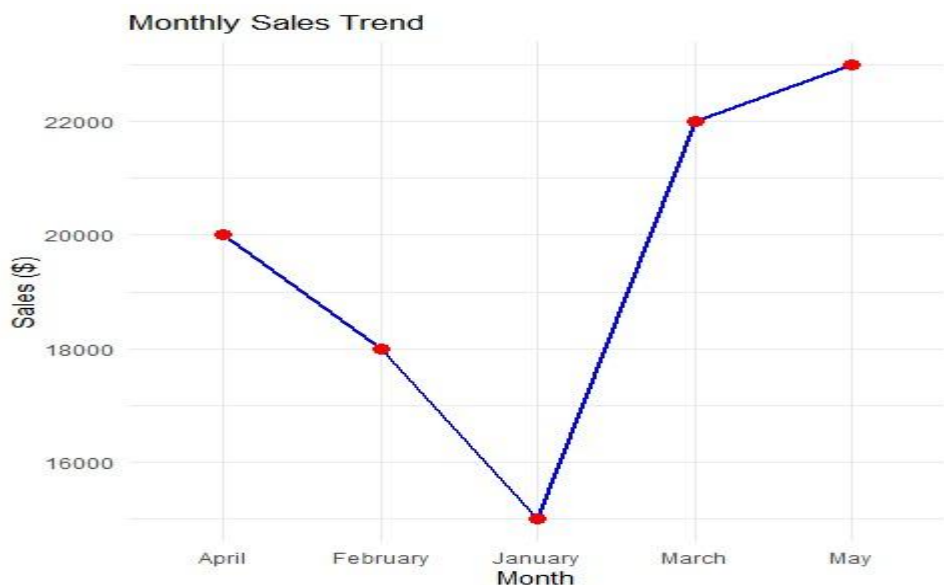


SET 18- Time Series Analysis.

18 A. Time Series Line Chart

Program:

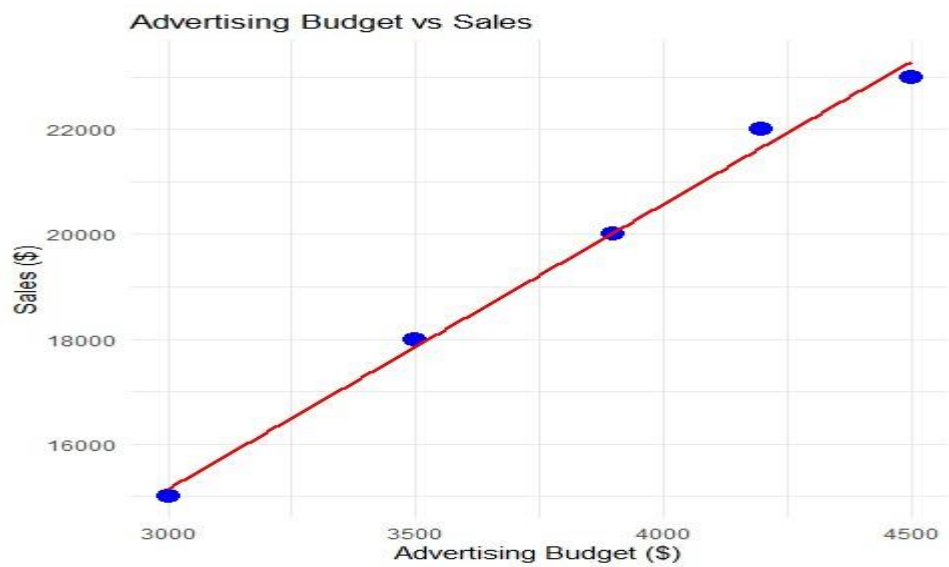
```
sales_data <- data.frame(  
  Month=c("January","February","March","April"  
  ,"May"),  
  Sales=c(15000,18000,22000,20000,23000))  
ggplot(sales_data, aes(x=Month, y=Sales,  
  group=1))+  
  geom_line(color="blue",linewidth=1)+  
  geom_point(color="red",size=3)+  
  labs(title="Monthly Sales Trend", x="Month",  
  y="Sales ($)")+ theme_minimal() Output:
```



18 B. Scatter Plot

Program:

```
sales_data$Advertising_Budget <- c(3000,3500,4200,3900,4500) ggplot(sales_data,  
  aes(x=Advertising_Budget, y=Sales))+ geom_point(size=4,color="blue")+  
  geom_smooth(method="lm",se=FALSE,color="red")+ labs(title="Advertising Budget vs  
  Sales", x="Advertising Budget ($)", y="Sales ($)")+  
  theme_minimal() Output:
```



18 C. Tableau Dashboard

Program:

```
line_plot <- ggplotly(
  ggplot(sales_data,
    aes(x=Month, y=Sales,
      group=1))+
    geom_line(color="blue")
  +
    geom_point(color="red",
      size=3)+
    labs(title="Sales
Trend")+
    theme_minimal())

pie_plot <-
  plot_ly(
    sales_data,
    labels=~Month,
    values=~Sales,
    type="pie"
  ) %>%
```

```
layout(title="Sales  
Distribution") dashboard
```

```
<- subplot( line_plot,  
pie_plot, nrow=1,  
shareX=FALSE,  
shareY=FALSE)  
dashboard
```

Output:

